

Directions: Answer each question in the space provided. Close book and closed notes. No electronic devices.

1. (30 points) Short Answer. You do not need to show your work for problems on this page.

(a) State two different (non-isomorphic) groups of order 9.

(b) State (in set notation) the cosets of $H = \langle 3 \rangle \subseteq \mathbb{Z}_9$. ($\mathbb{Z}_9$ is abelian, so you may give just the left cosets.)

(c) Write an addition table for $\mathbb{Z}_2 \times \mathbb{Z}_2$.

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(d) Give an example of a finite group that has more subgroups than it has elements.

(e) Let g be a generator of a finite cyclic group G , and let $H = \langle g^m \rangle \cap \langle g^n \rangle$. Give a formula for $|H|$.

(f) What is the smallest n for which A_n is non-abelian?

(g) Give an example of a permutation $\mu \in S_{12}$ for which $|\mu| = 28$.

(h) Is the permutation μ from part (g) even or odd?

(i) State the order of the permutation $(1\ 2\ 3\ 7\ 6)^{-1}(5\ 4) \in S_7$.

(j) Write $(1\ 2\ 3\ 4)^2(3\ 5)$ as a product of disjoint cycles.

2. (10 points) Does S_4 have any subgroups of order 10? If so, then state one. If not, explain why not.

3. (10 points) Does S_4 have any subgroups of order 4? If so, then state one. If not, explain why not.

4. (10 points) Prove or disprove: If every proper subgroup of a group G is cyclic, then G is abelian.

5. (10 points) Recall that the unit circle $\mathbb{T}$ is a subgroup of $\mathbb{C}^*$. Describe the cosets of $\mathbb{T}$ in $\mathbb{C}^*$. Justify.

6. (10 points) Suppose g is a fixed element of a group G .

Prove that the set $H = \{x \in G : xg = gx\}$ is a subgroup of G .

7. (10 points) Suppose p and q are distinct prime numbers. How many generators does $\mathbb{Z}_{pq}$ have? Explain.

8. (10 points) Prove: If G is a group and $(ab)^2 = a^2b^2$ for any two $a, b \in G$, then G is abelian.